Differences observed

When the data is sorted **Binary search tree** becomes like a linked list which make the search(), insert(), delete() time complexities **O(n) not O(log(n)).**

While **AVL** guarantees balancing which makes search(), insert(), delete() **always O(log(n))** but insert() and delete() requires more implementation (rotations).

When to use BST vs AVL?

Use **BST** when the **data is Random**.

Use **AVL** when the **data is Sorted**.

Time Complexities:

BST time complexities:

Method	**Avg (random data)**	**Worst (sorted data)**
search()	O(log(n))	O(n)
insert()	O(log(n))	O(n)
delete()	O(log(n))	O(n)

AVL time complexities:

Method	**Avg**	**Worst**
search()	O(log(n))	O(log(n))
insert()	O(log(n))	O(log(n))
delete()	O(log(n))	O(log(n))